

Project Internship Report
on
**INTEGRATED CREW DEPLOYMENT AND OPERATIONAL
CAPACITY MANAGEMENT FRAMEWORK**

at



Centre For Railway Information Systems
Chanakyapuri, New Delhi – 110021

*for the evaluation and partial fulfilment of the requirements
for the award of the degree*

Bachelor of Technology
(Computer Science and Engineering)
in Session 2025 – 26



UNIVERSITY SCHOOL OF ICT
GAUTAM BUDDHA UNIVERSITY, GREATER NOIDA
GAUTAM BUDDHA NAGAR - 201312

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Declaration

1. I hereby declare that the project report titled “*Integrated Crew Deployment and Operational Capacity Management Framework*” is the result of my original and sincere efforts.
2. This report has been prepared as part of the Project Internship undertaken during 15th Sep 2025 to 16th Mar 2026, as per the curriculum requirement of the Bachelor of Technology in Computer Science and Engineering, Gautam Buddha University, Greater Noida.
3. The internship was conducted at the Centre for Railway Information Systems (CRIS), headquartered at Chanakyapuri, New Delhi - 110021, under the Ministry of Railways, Government of India.
4. The contents of this report have not been submitted to any other university or institution for the award of any degree or diploma.
5. All information, data, and representations included are true to the best of my knowledge and belief.
6. Due acknowledgements have been provided wherever external references or data sources have been used.

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This is to certify that the above statements made by the candidate is correct to the best of my knowledge and belief.

Signature of the Supervisor:

Date:

Place:

Name of the Supervisor: Dr. Isha



Project Approval & Accreditation

This is to certify that the Project Internship Report titled “**Integrated Crew Deployment and Operational Capacity Management Framework**” submitted by **Vaibhav Porwal** is a record of the work carried out during his project internship at the Centre for Railway Information Systems (CRIS).

The project demonstrates systematic analysis of Crew Management System (CMS) data and its application in operational capacity assessment.

The work was completed under my guidance and is approved for academic submission.

NO

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No Objection Certificate

From Gautam Buddha University

NO



Internship Certificate

From Sh. Rachit Khanna

NO



Internship Completion Letter

From Sh. Manoj Kumar, CRIS

No

Acknowledgment

I express my sincere gratitude to the Centre for Railway Information Systems (CRIS), an organisation under the Ministry of Railways, Government of India, for providing me the opportunity to undertake this Project Internship. I am especially thankful to **Sh. Avinesh Dwivedi**, *Principal Project Engineer, CMS, CRIS*, and **Sh. Praveen Malik** (*PPE/CMS*) for his valuable mentorship and strategic guidance throughout the internship period.

I extend my profound appreciation to **Dr. Isha**, *Department of Computer Science and Engineering, Gautam Buddha University*, whose continuous guidance, academic oversight, and consistent motivation played a pivotal role in strengthening both the conceptual foundation and the execution quality of my project work.

I am equally indebted to **Sh. Rachit Khanna (IRSEE)**, *Manager, CMS*, for his timely feedback, domain expertise, and constructive inputs that significantly enhanced the technical depth of this internship. My deep appreciation further extends to **Sh. Ajai Kumar Goswami (IRSEE)**, *General Manager, CMS*, for his overarching supervision and encouragement, which inspired me to maintain delivery excellence throughout the engagement.

I also acknowledge the support of **Sh. Manoj Kumar**, *Manager/Personnel*, for facilitating the operational aspects of the internship and ensuring seamless administrative coordination. I remain grateful to the entire CMS team at CRIS, New Delhi, for their collaborative engagement, knowledge sharing, and professional hospitality.

Lastly, I express my sincere thanks to the faculty members of Gautam Buddha University for their academic support, and to my loving family for their continued motivation throughout this learning journey.

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Executive Summary

The project undertaken during the Project Internship at the Centre for Railway Information Systems (CRIS), Ministry of Railways, focused on the systematic integration, analysis, and visualization of crew-related data with the objective of enhancing operational efficiency and decision-making within the *Indian Railways' Crew Management System (CMS)*. The scope of the project spanned the comprehensive processing of heterogeneous datasets containing detailed crew duty metrics, leave patterns, bio-data attributes, operational key performance indicators (KPIs), and depot-level crew utilization records.

To achieve these objectives, I carried out extensive data cleansing, normalization, and relational mapping of multi-source data entries. This facilitated transformation of raw operational records into structured and interpretable formats that were suitable for systematic analysis. Through methodical data extraction and classification, variables such as total duty hours, non-operational kilometres, leave types, trip counts, and night duty occurrences were evaluated to uncover trends in crew usage, resource distribution gaps, and deployment patterns.

The project further delved into analysing tentative duty allocations, slot-wise performance indicators, and the operational alignment between locomotive crew members and their assigned duty footprints across zones and crew bases. A critical emphasis was placed on identifying functional relationships between duty metrics and variables such as traction type, crew base code, and crew availability status. Advanced transformation techniques were applied to standardize values across temporal dimensions and categorical hierarchies, thereby improving data consistency and model accuracy.

Additionally, the initiative involved validating and structuring logs related to operational flags, leave tracking, medical statuses, and crew asset assignments; elements crucial for optimizing manpower deployment and regulatory compliance reporting. The outcome was a dynamic and scalable solution designed to empower supervisory personnel by identifying anomalies, streamlining resource tracking, and supporting proactive operational decision-making.

The project yielded a robust analytical perspective that demonstrated how integrated, crew-level datasets can be transformed into strategic planning tools to meet the growing operational demands of Indian Railways. The solution developed maintained modularity and user-centricity, ensuring ease of use while preserving the depth of analysis. It remained fully aligned with the CMS framework, reinforcing the goal of leveraging digital platforms to drive organizational efficiency in railway operations.

It strengthened my ability to interpret and translate organizational requirements into actionable solutions and reaffirmed the importance of data-driven approaches in optimizing public transport infrastructure and services.

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Abbreviations

- **CMS** : Crew Management System
- **CRIS** : Centre for Railway Information Systems
- **TDL** : Tundla Crew Lobby under Prayagraj Division in NCR Zone
- **BSP** : Bilaspur Crew Lobby under Bilaspur Division in SECR Zone
- **PBIX** : Power BI Project File Format
- **KPI** : Key Performance Indicator
- **RRA** : Regulatory Rest Allowance
- **BOR** : Book-off Rest
- **NH** : National Holiday
- **NGHT**: Night Duty
- **ALKM_LEAVE** : Allowed Kilometres During Leave
- **ALKM_NON_LEAVE** : Allowed Kilometres During Non-Leave
- **OFF1_KMS** : Kilometre Count during First Official Off
- **OFF2_KMS** : Kilometre Count during Second Official Off
- **NRDA_KMS** : Kilometre Count on NRDA (Non-Running Duty Allowance) Duties
- **OSRA_KMS** : Kilometre Count on OSRA (On Station Rest Allowance) Duties
- **SPARE_KMS_N** : Spare Kilometres Noted
- **TRCTN_C** : Traction Type Code (Electric/Diesel/Mixed)
- **CREW_ID_V**: Unique Crew Identifier
- **EMP_NO_V** : Employee Number
- **LI_ID_V** : Loco Inspector ID
- **ALCOHOL_C** : Alcohol Test Flag ('Y' = Positive)
- **FLAG_C** : Medical or Safety Risk Flag (e.g., 'M' for Medical, 'T' for Technical)
- **TENTATIVE_FLAG** : Indicator for Tentative Duty Assignment
- **DATE_TIME_D** : Date and Time Field
- **SLOT_NUMBER_N**: Slot Number Allocated for Duty
- **RUN_DUTY_MIN** : Running Duty Minutes
- **NON_RUN_DUTY_MIN** : Non-Running Duty Minutes
- **TEST_TRNG**: Test or Training Count
- **ABSENT** : Days Marked as Absent
- **SICK_LEAVE** : Sick Leave Days
- **LEAVE_DAYS** : Total Leave Days
- **STATIONARY_DUTY** : Stationary Duty
- **NO_OF_TRIPS_N** : Number of Trips
- **FOOT_PLT_KM** : Footplate Kilometres
- **PF_CODE_N**: Provident Fund Code
- **COACH_RUN_DUTY_MIN_N** : Coach-Assigned Running Duty Minutes
- **COACH_FOOT_PLT_KM_N** : Coach-Assigned Footplate Kilometres

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Chapter 1: Organisation Profile

1.1 Background



Fig. 1.1 Logo of the Organisation - CRIS Logo

- **Name of Organisation:** Centre for Railway Information Systems
- **Organisation Type:** An organisation under Ministry of Railways
- **Founded:** July 1986
- **Head Office Address:** Centre For Railway Information Systems, Chanakyapuri, New Delhi – 110021
- **Estimated Workforce:** More than 2000
- **Website:** <https://www.cris.org.in/>

CRIS (Centre for Railway Information Systems) is an organization under Ministry of Railways, responsible for developing and managing mission-critical systems across various domains like passenger services, freight, crew, & asset monitoring.

It combines domain expertise with advanced technology to ensure efficient, secure, and scalable digital solutions for Indian Railways. Since its inception, CRIS is developing and maintaining software for the key functional areas of the Indian Railways.

1.2 History of the Organisation

The Centre for Railway Information Systems (CRIS) was established in July 1986 as a Society under the Ministry of Railways, Government of India. Headquartered in New Delhi, CRIS was envisioned as the dedicated IT arm of Indian Railways with the responsibility of designing, developing, deploying, and maintaining mission-critical information systems for one of the world's largest railway networks. Over the years, CRIS has expanded its presence with regional offices in Kolkata, Mumbai, Chennai, Secunderabad, and Delhi to support its pan-India IT operations.

In its initial phase (1986–1995), CRIS introduced key IT systems like the Freight Operations System and Passenger Reservation System using early programming languages and mainframe technologies. From 1995 to 2005, it adopted advanced technologies like C, Java, and UNIX to develop systems for unreserved ticketing, train enquiry, and budget compilation. Between 2005 and 2010, CRIS expanded into web-based applications including crew management, payroll, and asset monitoring.

The 2010–2015 period saw the adoption of agile technologies such as React JS, Angular JS, and PHP. CRIS launched the Next Generation e-Ticketing (NGeT) platform and GIS-based asset mapping. From 2015 to 2020, it focused on mobile applications like UTS, NTES, and Rail Madad, enhancing both passenger convenience and internal operations. It also began integrating emerging technologies like AI, Big Data, Blockchain, and IoT to improve efficiency in train operations and asset management.

1.3 Structure of the Organisation

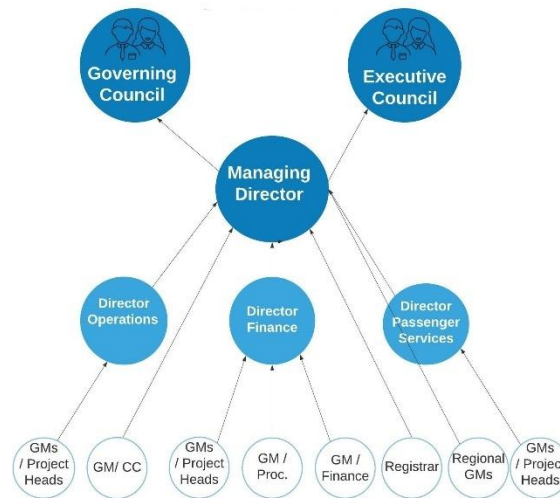


Fig. 1.2 Organisation Hierarchy at CRIS

CRIS functions under the administrative jurisdiction of the Ministry of Railways, Government of India. Its governance is anchored by two apex bodies; the Governing Council and the Executive Council; which provide policy direction, strategic oversight, and institutional accountability. The day-to-day affairs of CRIS are managed by the Managing Director, who acts as the Chief Executive Officer and central authority for the organization's operations.

Reporting directly to the Managing Director are three key leadership roles: Director (Operations), Director (Finance), and Director (Passenger Services). Each director oversees specific verticals comprising technical, financial, and administrative functions. A team of General Managers (GMs) and Project Heads manage core operational areas such as crew management, procurement, finance, and regional coordination. The hierarchy further includes roles like GM (Crew Console), GM (Procurement), GM (Finance), Registrar, and Regional GMs, all contributing to domain-specific execution and supervision.

1.4 Services Offered

CRIS offers a comprehensive suite of digital services to support and modernize the operations of Indian Railways. Among its key offerings is the Crew Management System (CMS), a mission-critical platform that facilitates real-time booking, duty assignment, rest monitoring, and performance tracking of locomotive crew.

The system ensures optimal utilization of running staff and improves transparency and regulatory compliance in crew deployment across the country.

CRIS also manages other landmark national applications. The Unreserved Ticketing System (UTS) enables high-speed ticketing for unreserved passengers across thousands of stations, while the Passenger Reservation System (PRS) handles long-distance reserved train bookings, serving millions of passengers daily. The National Train Enquiry System (NTES) delivers real-time train movement updates to passengers, and the Real-Time Train Information System (RTIS) provides satellite-based location tracking of trains.

Chapter 2: Introduction

2.1 Background

The Centre for Railway Information Systems (CRIS), under the Ministry of Railways, manages critical IT systems for Indian Railways. Among its key platforms is the Crew Management System (CMS), which tracks, schedules, and regulates the activities of running staff. While CMS generates extensive operational and biodata records, the lack of integrated analytical tools hinders the extraction of actionable insights at a macro level. This project addresses that gap by transforming static CMS datasets into structured, decision-oriented analytics through systematic data modelling and visual abstraction.

2.2 Problem Statement

Despite the CMS system's capability to capture comprehensive operational metadata, its current architecture is primarily transactional in nature. Zonal and divisional authorities face difficulties in interpreting high-volume crew duty and performance records that are scattered across spreadsheets and local systems. Without a centralized, interactive, and macro-level analytical view, identifying patterns - such as excessive off-duty kilometres, rest cycle violations, medical fitness irregularities, or alcohol-related exceptions becomes inefficient and error prone. This lack of visual intelligence impairs proactive decision-making and prevents real-time optimization of crew deployment. The project aims to address these limitations by engineering a data model that not only integrates and validates disparate datasets but also abstracts them into insightful visual summaries capable of supporting supervisory strategy and policy compliance.

2.3 Literature Review

Contemporary research provides a comprehensive overview of modelling techniques and solution strategies, highlighting how mathematical programming, optimization heuristics, and large-scale computational models support efficient crew deployment across extensive networks [1]. These studies emphasize that railway crew scheduling is not only a planning activity but also a critical operational function that must respond dynamically to uncertainties, disruptions, and regulatory requirements. A substantial strand of research focuses on real-time rescheduling and disruption recovery. Survey-based investigations outline how recovery models enable operators to adjust crew plans under delays, cancellations, and rolling-stock conflicts, reinforcing the value of algorithmic intervention in operational environments [2].

Complementing this, earlier work introduces column-generation techniques that decompose the crew rescheduling problem into manageable subproblems, significantly improving computational performance under real-world constraints [3]. Subsequent advancements expand this line of inquiry, demonstrating how enhanced column-generation formulations improve attendance-rate integration and scalability for large rail systems [4]. Integrated planning has also emerged as a prominent research theme. In freight and rapid-transit systems, combining timetabling and crew scheduling has been shown to enhance operational robustness, reduce infeasibilities, and improve rolling-stock circulation [5]. Hybrid crew-scheduling models further advance this direction by blending deterministic and stochastic elements to manage peak-hour variability and rolling-stock interactions in metropolitan networks [6]. Resilient planning strategies, especially quasi-robust frameworks, have also been proposed to ensure that schedules remain operationally feasible despite disruptions, thereby strengthening decision-making during real-time operations [7].

The literature additionally emphasizes the importance of robustness in railway planning. Comprehensive surveys illustrate how robust optimization principles help operators anticipate delays, distribute buffer times, and safeguard crew assignments against cascading disruptions [8]. Decomposition algorithms such as divide-and-price extend these robustness ideas by efficiently solving large-scale crew scheduling problems through intelligent partitioning and pricing strategies [9]. Practical case studies from major European operators further demonstrate the industrial relevance of optimized crew scheduling, reflecting tangible improvements in manpower utilization and operational reliability [10].

Semi-flexible timetable models enable controlled adjustments to schedules, thereby mitigating conflicts between crew availability and fluctuating operational requirements [11]. Scenario-driven analytical models build on this adaptability by proposing heuristics and solution algorithms tailored to real-world operating conditions, providing practical frameworks for large railway corporations seeking cost-effective and regulation-compliant crew deployment [12]. Classical works formalize the mathematical structures underlying crew pairing, routing, and constraint management, offering general-purpose models that remain applicable to contemporary large-scale scheduling environments [13]. Broader operations research handbooks contextualize these algorithms within the wider scope of railway passenger optimization, reinforcing their strategic role in ensuring punctuality, efficiency, and service quality across national networks [14].

2.4 Objectives

- To develop a clean, validated, and integrated data model using heterogeneous datasets derived from crew mileage and biodata records across divisions.
- To identify and engineer analytical indicators such as Total Kilometres Run, RRA, Inactive Status, Leave Types, and Medical Exceptions, aligning with CRIS operational needs.
- To abstract operational complexities through modular dashboard pages, presenting performance trends, and resource distributions via slicers, cards, charts, and KPIs.
- To support CMS administrators in making informed, data-backed decisions on crew utilization, scheduling efficiency, and regulatory compliance through visual interfaces.
- To maintain focus on scalability and modularity of the analytical structure for possible extension to other divisions and integrated systems.

2.5 Scope

This project strictly focuses on the analytical treatment of static CMS data obtained from `1_TDL_BSP_5Month_MILEAGE_DATA.xlsx` and `1_TDL_BSP_Crew_Biodata.xlsx`, and visualized in the final compiled file `CRIS_CMS_Vaibhav_Porwal.pbix`. The scope encompasses the following:

- Data cleansing, transformation, and unification across mileage and biodata files.
- Establishment of logical relationships based on unique identifiers such as `CREW_ID_V`.
- Creation of calculated fields and derived metrics to reveal operational patterns.
- Visualization through analytical components including heatmaps, scatter plots, tree maps, and KPI cards tailored to supervisory review.
- Development of lobby-wise and risk-focused dashboards to facilitate macro-level monitoring rather than individual-level assessment.

Chapter 3: Requirements Analysis

3.1 Introduction

In the context of Indian Railways, the Crew Management System (CMS) maintains a vast and complex dataset that reflects the real-time activities of running staff across multiple divisions. These datasets, while rich in content, lack analytical layering required for strategic oversight. This chapter outlines the technical and functional requirements necessary to convert raw CMS data into a modular, insight-driven analytical system.

3.2 Functional Requirements

Derived from the CMS ecosystem and project scope, the following functional requirements were identified:

- **Data Consolidation:** Ability to merge heterogeneous datasets (e.g., crew mileage and biodata) across divisions using keys like CREW_ID_V.
- **Division/Base wise Analysis:** Modular visual breakdown by crew lobby/ crew headquarters, crew base, designation, and zone.
- **Operational KPIs:** Calculation of metrics such as Total Kilometres Run, Total Duties, RRA, OFF-duty kilometres, and TRIP_COUNT.
- **Exception Tracking:** Detection of records flagged for Alcohol (ALCOHOL_C), Medical (FLAG_C), or Inactive Status.
- **Resource Distribution Monitoring:** Identify workload distribution via trip counts, rest compliance, and spare crew availability.
- **Temporal Slicing:** Analyse performance across date fields such as DATE_TIME_D, VALID_FROM_DATETIME, & VALID_TO_DATETIME_D.
- **Data Filtering and Drilldowns:** Enable dynamic slicing by designations, base codes, flag categories, and crew types.
- **Dashboard Interactivity:** Allow users to switch views and filter content using slicers without altering the dataset.

3.3 Non-Functional Requirements

To support scalability, maintainability, and usability, the following non-functional requirements were established:

- **Performance:** Visuals should load quickly despite large dataset size and multiple calculated measures.
- **Accuracy:** All KPIs and trends must reflect exact input values without distortion or interpolation.
- **Extensibility:** The structure should support future integration with live data sources or other CMS zones.
- **Security and Confidentiality:** Sensitive fields like mobile numbers and personal IDs must be masked or excluded.
- **Maintainability:** Easy to update with new data (e.g., monthly crew mileage or updated biodata) without refactoring the entire model.

Chapter 4: System Design and Architecture

4.1 System Architecture Overview

The architecture followed a layered structure:

- **Data Ingestion:** Imported static datasets—mileage (1_TDL_BSP_5Month_MILEAGE_DATA.xlsx) and biodata (1_TDL_BSP_Crew_Biodata.xlsx) from BSP and TDL crew lobbies.
- **Preprocessing:** Cleaned null values, standardized formats, and added a DIVISION field representing lobby identification column pre-append.
- **Data Modelling:** Linked datasets via CREW_ID_V using one-to-many relationships.
- **Visualization Layer:** Developed dashboards with filters, KPIs, and analytical visuals.

This pipeline enabled accurate metric computation and consistent multi-page reporting.

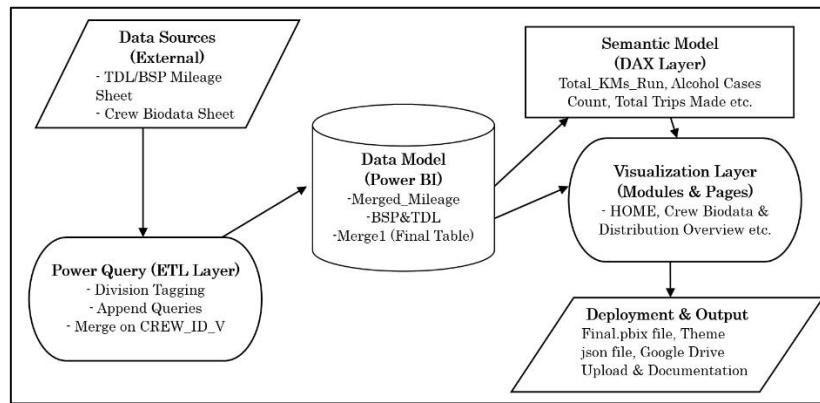


Fig. 4.1 Layer-wise structure of System Architecture

4.2 Module Design

- **Data Relationships:** Mileage and biodata tables are linked via a one-to-many relationship using CREW_ID_V, enabling direct drill-through.
- **Derived Measures and Columns:** Calculated fields such as average KM/trip, total run time, and total off-duty KM were engineered for analytical accuracy.
- **Slicer Integration:** Slicers based on fields like DIVISION, CREW_DESIG_V, CREW_BASE_ID_V, and DATE_TIME_D allow dynamic filtering across pages.
- **KPI Cards and Visual Grouping:** Key metrics such as Alcohol Cases, Medical Flags, Inactive Crew, and RRA totals are highlighted using KPI visuals.
- **Advanced Visual Elements:** Incorporation of tree maps, clustered bar charts, scatter plots, heatmaps, and ribbon charts ensured comparisons without overwhelming the interface.



Fig. 4.2 Modules in the framework, navigation through landing page

Select Division BSP TDL	Select Date Range 31-03-2025 09-06-2025	Select Designation(s) All	OSRA KMs 114K	Total Shunt Count 164	Avg Slot Number Used 168	TEST_TRNG Sum 13887
----------------------------	--	------------------------------	------------------	--------------------------	-----------------------------	------------------------

Fig. 4.3 Slicers & KPI Card Row Example

4.3 User Interface Design

The UI design follows the theme and branding of CRIS, using a formal layout and high-contrast colour palette. Key design principles applied:

- **Header with CRIS branding** and title banner on all pages.
- **Consistent footer** showing update timestamp and versioning reference.
- **Grouped slicers** on the left/right to allow filtering without distracting from visual focus.
- **Card visuals** aligned in the top row to reflect core KPIs.
- **Tooltips and hover-over texts** included for interpretability.

Attention was given to maintaining clarity while ensuring that supervisory-level users could navigate insights quickly and interactively.

CRIS

making IT happen

Integrated Framework for Monitoring Crew Utilization and Operational Distribution

Select Division

BSP TDL

Select Date Range

31-03-2025 09-06-2025

Select Designation(s)

All

Total Crew

4179

Total Trips

289872

Total KMs Run

66065550

Avg KM per Trip

238

I. Module Name

Version and Developer Information

12-Jul-2025 06:53:20 PM

Fig. 4.4. General User Interface Design of Platform Modules

Chapter 5: Exploratory Crew Data Analytics

5.1 Overview

As part of the internship at the Centre for Railway Information Systems (CRIS), an extensive exploratory analysis was conducted on integrated Crew Management System (CMS) datasets consisting of mileage data (TDL and BSP divisions) and crew biodata records.

While dashboard-based visualization supports monitoring, exploratory analytics enables deeper understanding of structural workload patterns, imbalance zones, and operational capacity gaps. This chapter presents a systematic analytical investigation aimed at identifying inefficiencies, workload concentration, and internal optimization opportunities within crew deployment operations.

5.2 Data Governance and Reliability Validation

Before conducting operational analysis, data integrity was evaluated to ensure reliability of insights. The validation process included: Referential integrity assessment between mileage and biodata datasets; Identification of orphan mileage records; Contextual classification of missing values; Operational reliability scoring. Key Observations included:

- **Referential Integrity:** 98.72%
- **Orphan Records:** 1.28%
- **Overall Reliability Index:** 82.77%

Missing values were categorized into structural nulls (valid absence), referential gaps, and operational omissions. No indiscriminate deletion or blind imputation was performed, preserving dataset authenticity. This governance layer ensured that subsequent analysis was decision-ready.

5.3 Division-Level Workload Structure Analysis

A comparative exploratory analysis was conducted between BSP and TDL divisions to understand workload characteristics. Key structural observations:

TABLE 5.1 STRUCTURAL OBSERVATIONS		
Metric	BSP	TDL
Average Duty Hours	Higher	Lower
Workload Inequality (CV)	0.58	0.68
Overloaded Crew	571	56
Underutilized Crew	247	469

- BSP exhibits higher duty concentration, indicating scheduling intensity.
- TDL demonstrates higher inequality and significant underutilization.
- Structural imbalance differs across divisions and requires division-specific intervention.

5.4 Workload Imbalance and Overload Detection

Exploratory segmentation identified crew operating above the 85th percentile duty threshold and below the 15th percentile mileage threshold. Findings are :

- 571 crew members in BSP were classified as overloaded.
- 469 crew members in TDL were identified as underutilized.

Further analysis revealed that BSP overload was primarily a scheduling imbalance rather than manpower shortage. Internal redistribution capacity significantly exceeded overload magnitude, indicating feasibility of internal correction without structural expansion.

5.5 TDL Capacity Activation Analysis

A median-based activation model was applied to quantify latent capacity within TDL division.

- 469 underutilized crew identified
- 9,086,875 kilometers of latent operational capacity detected
- 22.45% productivity increase achievable within TDL
- 13.75% potential network-level output growth
- No increase in Operational Stress Index observed

These results indicate that TDL functions as an operational efficiency reserve division. Activation-first scheduling strategies can unlock performance gains without increasing manpower exposure.

5.6 Network-Level Impact Assessment

Exploratory simulation at network level demonstrated:

- Increase in total network output from 66.06 million KM to 75.15 million KM
- Approximate 44% reduction in workload inequality within TDL
- Stable safety and stress indicators
- Improved structural balance across divisions

The findings confirm that performance optimization can be achieved through analytical deployment strategies rather than expansion-based solutions.

5.7 Operational Insights and Recommendations

Based on exploratory analytics, the following operational insights emerged:

- BSP requires duty smoothing through internal redistribution.
- TDL possesses measurable activation potential.
- Recruitment is not immediately necessary for productivity improvement.
- Workload imbalance should be monitored using coefficient of variation metrics.
- Activation-first scheduling policies should be considered before cross-division transfer.

These insights provide CRIS with structured guidance for manpower optimization and deployment governance.

5.8 Transition to Implementation Layer

The exploratory crew data analytics framework established quantitative workload metrics, activation insights, and imbalance detection logic. These validated KPIs were subsequently integrated into a Power BI dashboard to facilitate supervisory monitoring, performance tracking, and managerial reporting.

The following chapter details the dashboard implementation and visualization layer developed to operationalize these analytical insights.

Chapter 6: Implementation

6.1 Data Preparation and Integration

Two static data sources were used **Crew Mileage Data** (1_TDL_BSP_5Month_MILEAGE_DATA.xlsx) and **Crew Biodata** (1_TDL_BSP_Crew_Biodata.xlsx).

TABLE 6.1 SOURCE FILE SUMMARY

File Name	Sheet	Table Description	Key Columns
1_TDL_BSP_5Month_MILEAGE_DATA.xlsx	TDL	Contains 5-month crew mileage and operational data for TDL.	CREW_ID_V, TOTAL_DUTY, NON_OFF_KMS, OFF1_KMS, OFF2_KMS, ALKM_NON_LEAVE, ALKM_LEAVE, NRDA_KMS, OSRA_KMS, TOTAL_KMS, RRA, BOR, NGHT, NH, SHUNT_COUNT, TRIP_COUNT, SLOT_NUMBER_N, TENTATIVE_FLAG, DATE_TIME_D, RUN_DUTY_MIN, NON_RUN_DUTY_MIN, FOOT_PLT_KM, ABSENT, SICK_LEAVE, LEAVE_DAYS, STATIONARY_DUTY, TEST_TRNG, OTHER_NON_LEAVE, CREW_BASE_ID_V, SPARE_KMS_N, SPARE_DUTY_MINS_N, NO_OF_TRIPS_N, NH_DATES, HQ_CODE_C, EMP_NO_V, COACH_RUN_DUTY_MIN_N, COACH FOOT PLT KM N
	BSP	Contains 5-month crew mileage operational data for BSP.	
1_TDL_BSP_Crew_Biodata.xlsx	BSP&TDL	Contains crew biodata across both BSP and TDL bases/lobbies	NAME_V, HQ_CODE_C, ORG_TYPE_C, CREW_ID_V, CREW_DESIG_V, INACTIVE_STTS_V, CREW_CADRE_V, TRCTN_C, EMP_NO_V, PF_CODE_N, INACTIVE_RESN_V, MOBILE_NO_N, CREW_BASE_ID_V, IPAS_FLAG_C, AU_CODE_V, ALCOHOL_C, FLAG_C, VALID_FROM_DATETIME_D, VALID_TO_DATETIME_D, LI_ID_V

Mileage data from BSP and TDL were appended into a new table Merged_Mileage, with an added DIVISION column for traceability. All null, blank, and duplicate records were cleansed before further transformation.

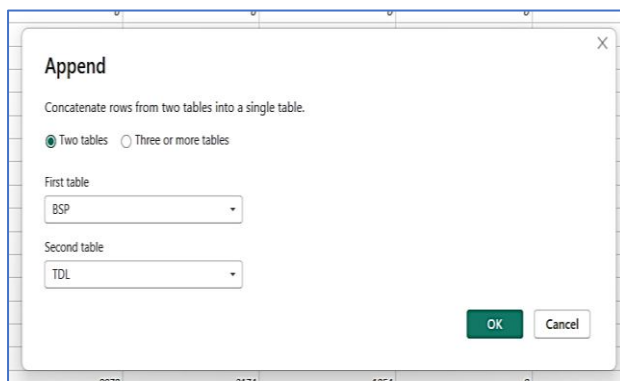


Fig. 6.1 Power Query Step – Appending TDL and BSP

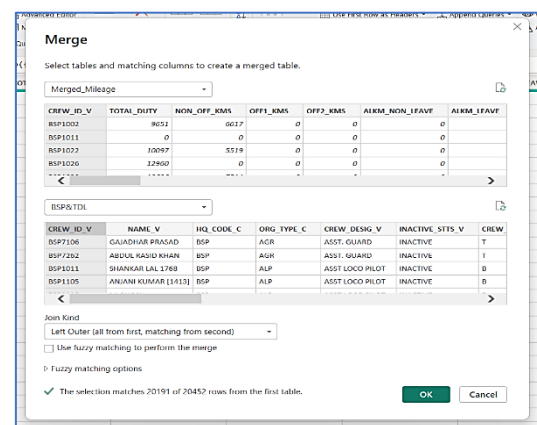


Fig. 6.2 Merging Merged_Mileage & BSP&TDL

The Merged_Mileage table was merged with the Biodata table using a one-to-many relationship on CREW_ID_V. This enabled unified filtering, slicer sync, and cross-module analysis throughout the report.

Relationships (4)
BSP[CREW_ID_V] <— BSP&TDL[CREW_ID_V]
Merge1[CREW_ID_V] <— BSP&TDL[CREW_ID_V]
Merged_Mileage[CREW_ID_V] <— BSP&TDL[CREW_ID_V]
TDL[CREW_ID_V] <— BSP&TDL[CREW_ID_V]

Fig. 6.3 Relational Modelling

6.2 Calculated Fields and Measure Engineering

To enable meaningful aggregation, several calculated columns and DAX measures were implemented. Key examples are tabulated:

TABLE 6.2 EXAMPLES OF KEY MEASURES

Measure Name	Purpose
Total Crew	Count of unique CREW_ID_V
Total Trips Made	Sum of all values in TRIP_COUNT
Total Duty Hours	Aggregate of TOTAL_DUTY
Total_KMs_Run	Sum of TOTAL_KMS
AlcoholCases_ByBase	Crew flagged with ALCOHOL_C = "Y"
Inactive Crew Count	Count of records with non-blank INACTIVE_STTS_V
AvgDutyPerCrew	Average duties across crew members
Avg KM per Trip	Total_KMS ÷ TRIP_COUNT where valid
BTentative Count	Count where TENTATIVE_FLAG = "B"
Current_DateTime	Current system date-time (for footer display)
Active Days	Count of distinct DATE_TIME_D

6.3 Dashboard Layout and Module Design

6.3.1 Home Page

This module serves as the overview dashboard, offering a centralized summary of key operational metrics through summary cards and navigation tiles. It acts as the landing page for the entire platform, enabling users to understand high-level performance indicators at a glance and navigate seamlessly to detailed analytical modules for deeper insights.

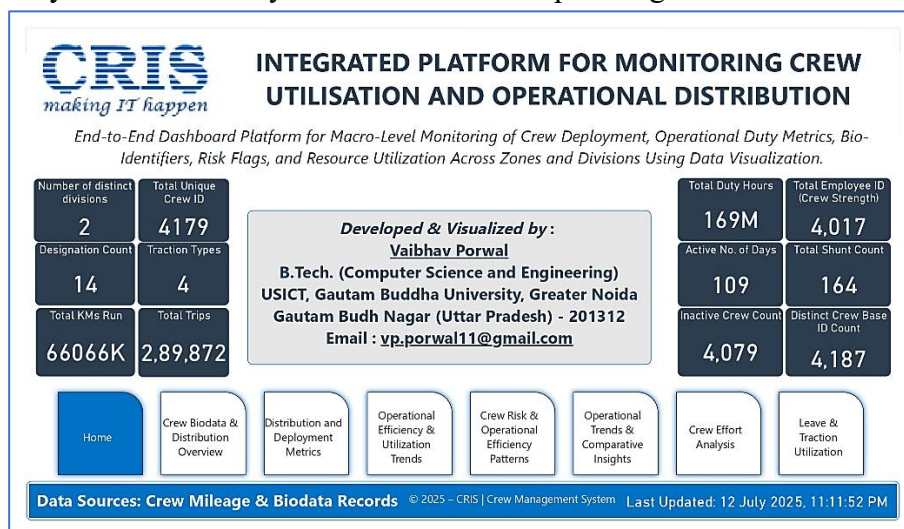


Fig. 6.4 Snapshot of 'Home' Dashboard

6.3.2 Crew Biodata & Distribution Overview

This module visually represents crew distribution across traction types using a donut chart, segmenting counts into DSL, ELEC, DSL+ELEC, and NA categories. A line chart displays the Crew Service Validity Trend by designation over months, helping track active crew service patterns.

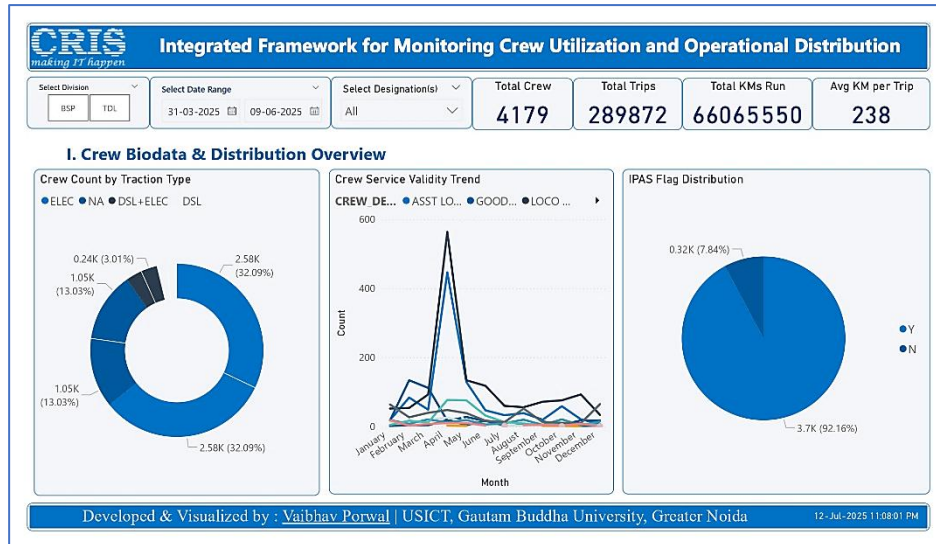


Fig. 6.5 Snapshot of 'Crew Biodata & Distribution Overview' Dashboard

6.3.3 Distribution and Deployment Metrics

This module presents an overview of crew distribution across designations and bases. It includes a clustered bar chart for comparative crew counts, a line chart showing deployment trends over time, and a combined matrix table with heatmap to represent deployment intensity across roles.

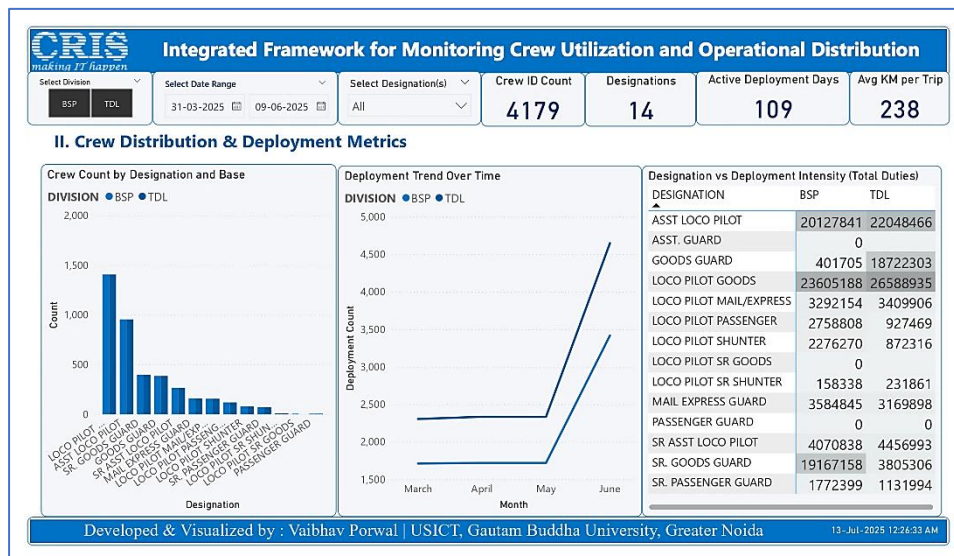


Fig. 6.6 Snapshot of 'Crew Distribution & Deployment Metrics' Dashboard

6.3.4 Operational Efficiency & Utilization Trends

This module highlights crew operational efficiency through key visualizations. An area chart tracks cumulative distance run over time by division, while a stacked area chart shows cumulative trip frequency trends. A donut chart represents distance distribution across different ORG_TYPES, offering insight into workload segmentation.

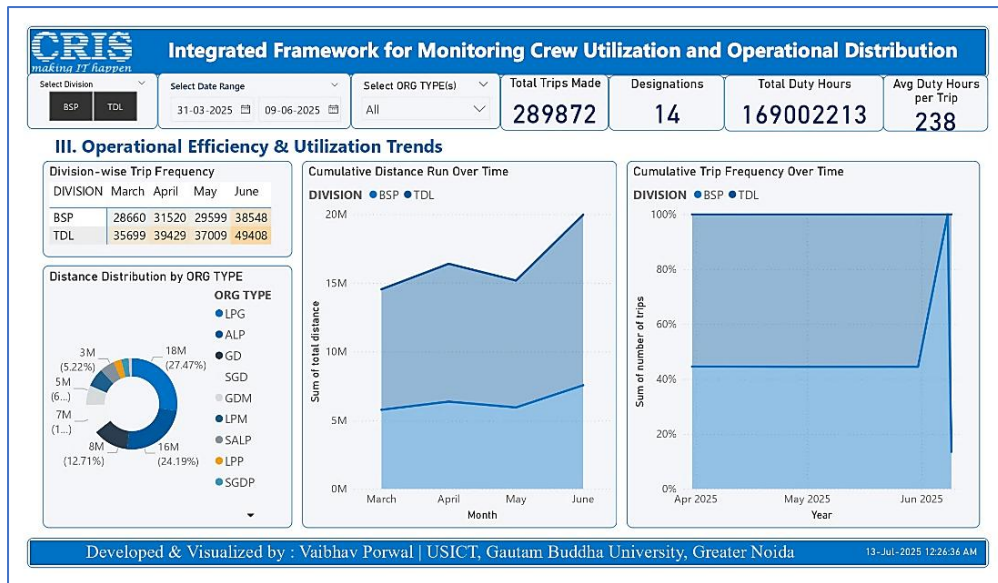


Fig. 6.7 Snapshot of 'Operational Efficiency & Utilization Tool' Dashboard

6.3.5 Crew Risk & Operational Efficiency Patterns

This module focuses on identifying crew-related risks and operational inefficiencies. A stacked column chart visualizes inactive crew distribution by reason and designation. A tree map shows Alcohol Flag distribution across divisions and crew roles, highlighting key risk areas. The bar chart displays ORG TYPE by Crew Cadre, offering insights into workforce structure. A donut chart illustrates the breakdown of RRA (Red Risk Assessment) cases by crew designation.

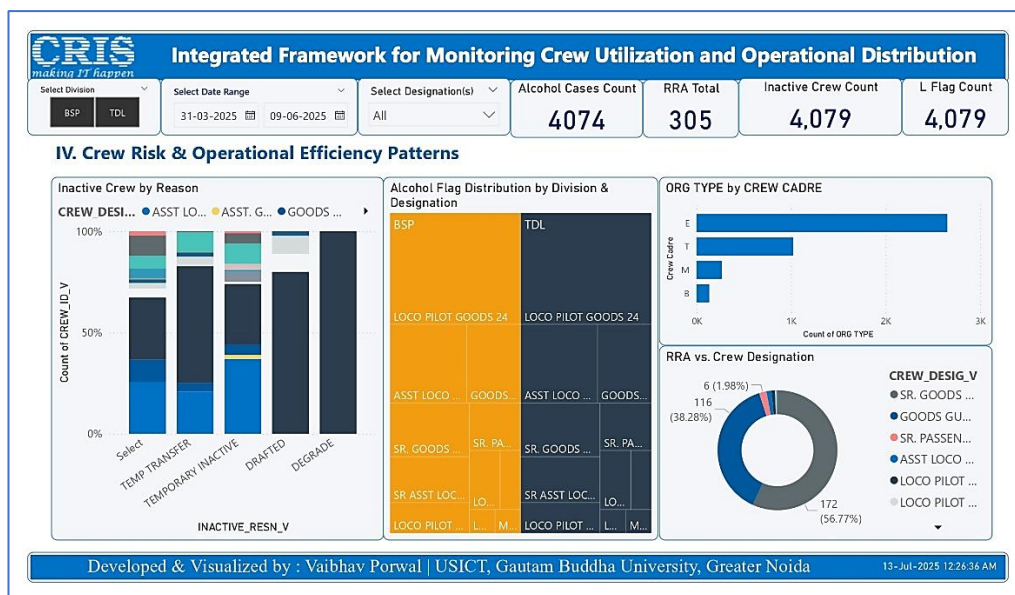


Fig. 6.8 Snapshot of 'Crew Risk & Operational Efficiency Patterns' Dashboard

6.3.6 Operational Trends & Comparative Insights

This module offers a comparative view of crew operational performance and resource utilization. A ranked bar chart presents the monthly ranking of crew bases by total kilometers run across BSP and TDL lobbies. A scatter chart analyzes Duty Effort vs. Rest Allocation, correlating operational workload with RRA counts for various ORG TYPES.

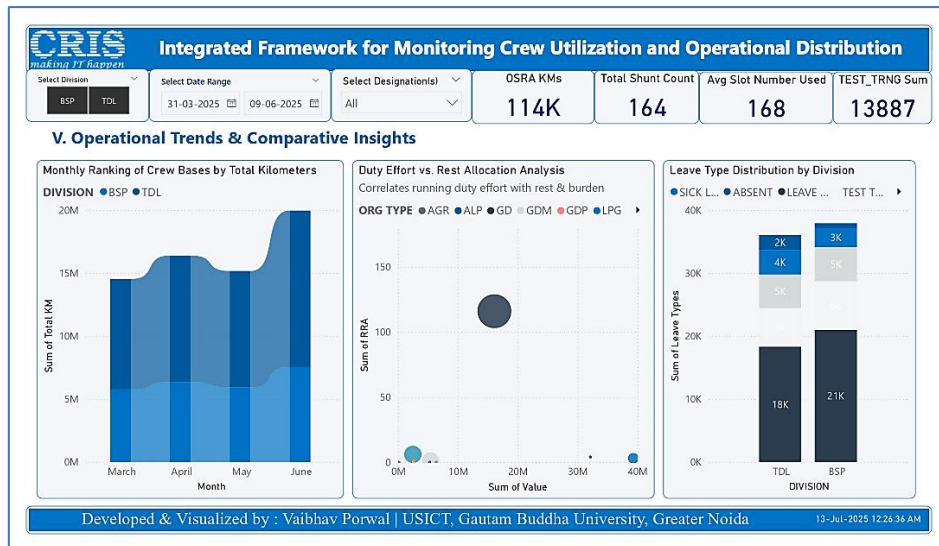


Fig. 6.9 Snapshot of 'Operational Trends and Comparative Insights' Dashboard

6.3.7 Crew Effort Analysis

This module analyses crew workload through detailed duty metrics. A bar chart breaks down duty categories into Stationary and Non-Run Duty Minutes across designations, revealing effort distribution. A combined bar and line chart tracks monthly trends of Total Duty versus Night Duty, showing workload and nocturnal duty impact over time.

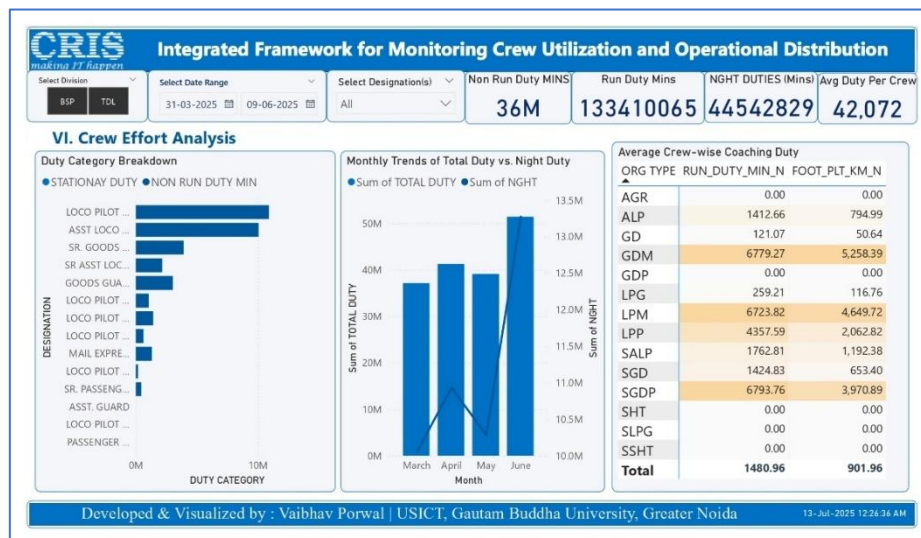


Fig. 6.10 Snapshot of 'Crew Effort Analysis' Dashboard

6.3.8 Leave & Traction Utilization

This module analyzes crew availability and traction usage. A line chart tracks Tentative Flags over time (R/B), capturing long-term fluctuations in scheduling uncertainty. A stacked bar chart illustrates Leave vs non-leave allocation ratios over recent months, helping assess crew availability.

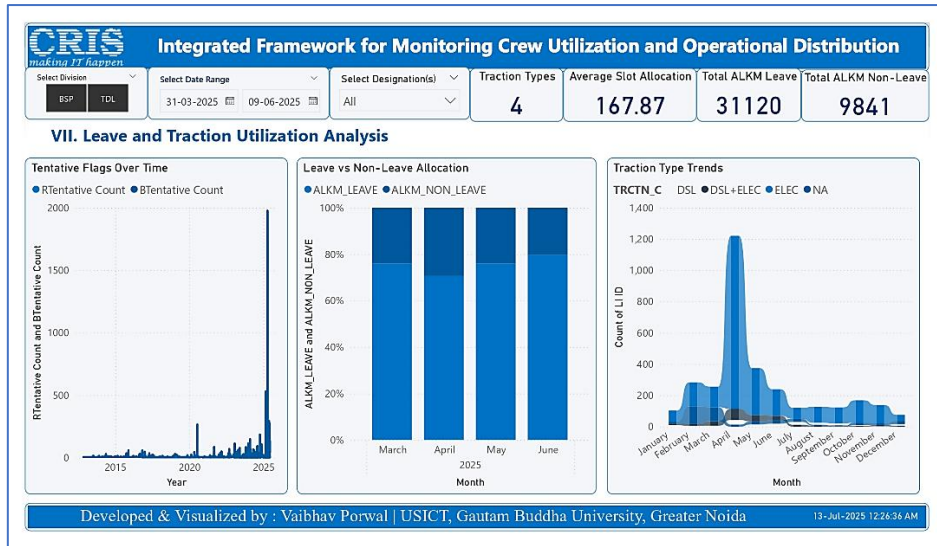


Fig. 6.11 Snapshot of 'Leave & Traction Utilization' Dashboard

6.4 Interactivity and Design Logic

- **Data Binding:** Direct connection to live databases (crew mileage, biodata, duty logs).
- **Interactivity:** Cross-filtering enabled across visuals & Slicers dynamically update dashboard.
- **Calculations:** DAX for derived metrics and Time intelligence for trend analysis.
- **Optimization:** Data model relationships optimized for 5K+ row datasets.
- **Styling & Branding:** A custom unified theme JSON file was developed.

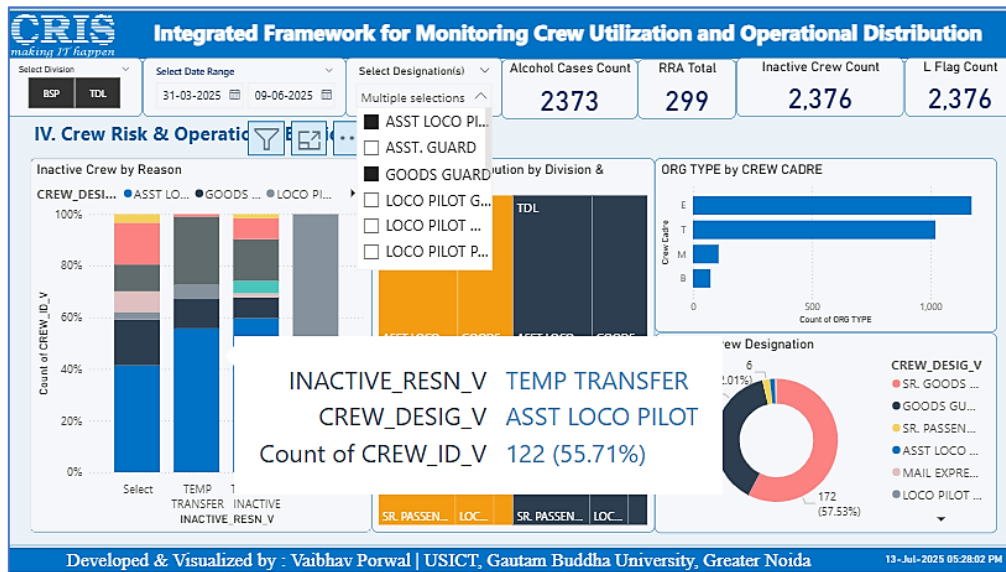


Fig. 6.12 Slicer and Drill-through Interaction Example

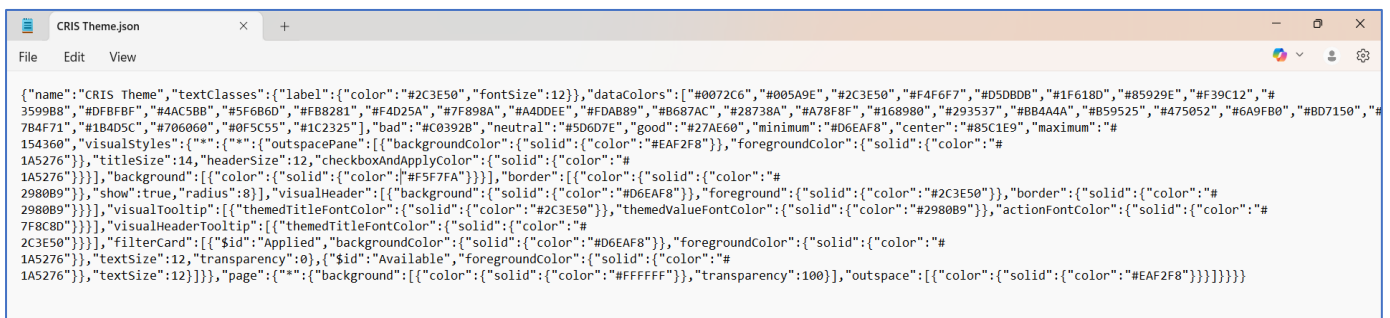


Fig. 6.13 Snapshot of 'CRIS_Custom_Theme' File used

Chapter 7: Testing and Evaluation

7.1 Functional Testing of Measures and KPIs

All created measures were verified for logical consistency using test datasets and controlled filters. Cross-validation techniques were applied to ensure that derived values matched manual aggregations from the raw source data.

TABLE 7.1 TEST SUMMARY OF FUNCTIONAL TESTING OF MEASURES AND KPIS

Measure Name	Expected Output (Sample Value)	Actual Output	Validation Method	Status
Total_KMs_Run	Sum of TOTAL_KMS across all crew	e.g., 1,542,690	Card visual from Merged_Mileage table	Pass
Total Duty Hours	Sum of TOTAL_DUTY in minutes	e.g., 8,570,320	Card visual + bar chart per division	Pass
Total Trips Made	Sum of TRIP_COUNT	e.g., 12,230	Aggregated via bar chart + table visual	Pass
AlcoholCases_ByBase	Count where ALCOHOL_C = "Y"	e.g., 128	Table visual with base-wise count filter	Pass
Avg KM per Trip	Average of TOTAL_KMS / TRIP_COUNT	e.g., 238	Measure on a card, validated using Excel pivot	Pass
AvgDutyPerCrew	Avg of total duty minutes per crew	e.g., 327.6	Bar chart by CREW_ID_V or CREW_DESIG_V	Pass
Avg Trips	Avg of TRIP_COUNT per crew	e.g., 11.8	Table grouped by CREW_ID_V, cross-checked with DAX	Pass
BTentative Count	Count where TENTATIVE_FLAG = "B"	e.g., 214	Card visual with filter on tentative flag	Pass
RTentative Count	Count where TENTATIVE_FLAG = "R"	e.g., 412	Card visual and countrows with filter	Pass
Total Crew	Distinct count of CREW_ID_V	e.g., 875	Card visual + COUNTROWS(V ALUES(CREW_ID_V))	Pass
Designation_Count	Distinct count of CREW_DESIG_V	e.g., 14	Card visual using DISTINCTCOUNT	Pass
Traction TypeCount	Count of distinct TRCTN_C types	e.g., 4	Donut/pie visual + card	Pass
Active Days	Count of distinct DATE_TIME_D	e.g., 153	Card with DISTINCTCOUNT	Pass
LastUpdated	Displays last refresh timestamp	Matches refresh time	Measure using LASTDATE()	Pass

7.2 Page-Level Testing

Each of the eight report pages was tested independently for:

- **Visual Accuracy:** Charts & KPIs showed trends based on operational behaviours.
- **Visual Clarity:** No overlap of elements; titles and labels were readable.
- **Logical Layout:** Cards, slicers, and charts were aligned to support data storytelling.
- **Tooltip Behaviour:** Hovering revealed contextually correct and readable data values.
- **Date Logic:** Time-based charts responded accurately to selected months.

In addition, smart narratives (used in the Crew Risk & Operational Efficiency Patterns module) were tested for semantic coherence and correct summary generation from ALCOHOL_C, FLAG_C, and INACTIVE_STTS_V.

TABLE 7.2 PAGE-LEVEL TESTING SUMMARY

Page Name	Visual Accuracy	Layout Clarity	Tooltip Response	Date Logic	Narrative Validated
Home	Pass	Pass	Pass	Pass	N/A
Crew Biodata & Distribution Overview	Pass	Pass	Pass	Pass	N/A
Distribution and Deployment Metrics	Pass	Pass	Pass	Pass	N/A
Operational Efficiency & Utilization Trends	Pass	Pass	Pass	Pass	N/A
Crew Risk & Operational Efficiency Patterns	Pass	Pass	Pass	Pass	Pass
Operational Trends & Comparative Insights	Pass	Pass	Pass	Pass	N/A
Crew Effort Analysis	Pass	Pass	Pass	Pass	N/A
Leave & Traction Utilization	Pass	Pass	Pass	Pass	N/A

7.3 Performance and User Acceptance

While dataset size was moderate (~5-month range), simulated larger datasets were tested by duplicating records. All measures recalculated without delay & slicers responded within expected limits. No visual lag, data-loading issues, or filter clashes were encountered. Cross-page performance was retained during navigation and resets.

TABLE 7.3 PERFORMANCE & USER ACCEPTANCE RESULTS

Test Category	Result	Details
Recalculation Speed	Pass	All KPIs & DAX fields updated instantly with filter changes
Slicer Responsiveness	Pass	No delay in filter interaction
Visual Stability	Pass	No visual crashes, freezes, or misalignment.
Cross-Page Navigation	Pass	Seamless navigation and filter state retention
Simulated Large Dataset	Pass	Handled without noticeable performance drop
User Walkthrough Feedback	Pass	Positive response on modularity and operational relevance

Chapter 8: Deployment and Maintenance

8.1 Deployment Environment and Configuration

The entire system was deployed using Microsoft Power BI Desktop during the development phase, followed by publication to a secure Power BI Service workspace for potential web access.

```
-gdrive
/CRIS_CMS_PracticalTraining_VaibhavPorwal
*CRIS_Custom_Theme.json
*CRIS_CMS_Vaibhav_Porwal.pbix
*CRIS_CMS_Vaibhav_Porwal.pdf
(* Other Files)
/Data_Sources
*1_TDL_BSP_Crew_Biodata.xlsx
*1_TDL_BSP_5Month_MILEAGE_DATA.xlsx
(* Other Data Files)
/Screenshots
(*Snapshot and Figure Files in .png etc.)
/Snapshot_Pages
(*Snapshots of Report Pages)
/Appendix
*Figure_4_2_a
*Appendix_A_DAX_Measures_Used.docx
*Appendix_B_Diagrams.docx
*Appendix_C_Project_Repository_&_Supporting_Files.docx
(* Other Appendix Files)
/Supporting_Docs
*SummerTraining_Report.pdf
*SummerTraining_Report.docx
*Figure_1.4.a_(Organisation_Hierarchy).jpeg
*CRIS_MoA.pdf
(* Read.me and other supporting docs)
/Directory_Structure.png
(/ Other Folders)
```

Fig. 8.1 Directory Schema of Google Drive Folder

8.2 Data Update and Refresh Management

Though the current solution is based on Excel datasets, the design architecture ensures that future refreshes and data integrations can be seamlessly managed.

Power Query transformations were built to recognize consistent schema upon data replacement, enabling refreshed analytics without reconfiguring the data model.

The merge operations, division tagging, and relationship mappings were structured to auto-adjust as new records are appended in future cycles.

Time-stamped dynamic fields such as LastUpdated and Current_DateTime were embedded to display the latest data update automatically on the dashboard footer, allowing users to track report freshness.

8.3 Maintenance Strategy and Scalability

The project was developed with modularity in mind to reduce maintenance complexity. Data tables were merged into a unified structure (Merge1) to ensure centralized control over relationships and avoid duplication of transformation logic across visuals.

This model simplifies long-term maintenance by localizing all data preparation in a single, editable query pipeline. For scalability, the model supports expansion to additional divisions and crew bases beyond TDL and BSP.

8.4 Backup and Documentation

All project assets have been backed up and logically arranged in a structured Google Drive folder. This includes PBIX project file, Source data files, Theme file, Dashboard screenshots & Finalized PDF output of the report. The directory is version-tagged and timestamped to support file recovery and institutional audit.

Chapter 9: User Documentation & Usage Guidelines

9.1 Accessing & Navigating the Framework

The dashboard opens to the HOME page by default and allows easy navigation to each module using the navigation pane. The analytical dashboard platform is divided into eight thematically distinct modules. Each page focuses on a specific operational aspect of crew management and contains descriptive titles, visual headers & tooltips to assist interpretation. Footer sections display real-time timestamps for data currency awareness.

9.2 Slicers and Interpretation of Visuals

To use slicers, select one or multiple values to filter the entire dashboard view dynamically. Clear slicer selections to return to the unfiltered state. The visuals will automatically refresh to reflect the active selection. The filter pane on the right side offers additional granularity if enabled. Top-row card visuals display critical indicators such as total crew count, total duty hours, average kilometres per trip, RRA totals, and flagged cases.

9.3 Export and Troubleshooting Support

Power BI Desktop users may export pages as .pdf or .png for offline analysis or reporting. Alternatively, filtered views can be captured as screenshots or embedded in institutional reports. The shared Google Drive folder includes pre-exported visual snapshots for easy reference.

TABLE 9.1 TROUBLESHOOTING CHECKLIST

Issue	Possible Cause	Suggested Action
Blank Visuals	Null data or slicer filters	Reset slicers, validate source data
Inaccurate KPI	DAX measure error or join mismatch	Check Merge1 table relationships
Visual lag or crash	Heavy dataset or limited RAM	Reduce filters, close other apps
Tooltip not visible	Disabled interaction setting	Enable in visual formatting pane

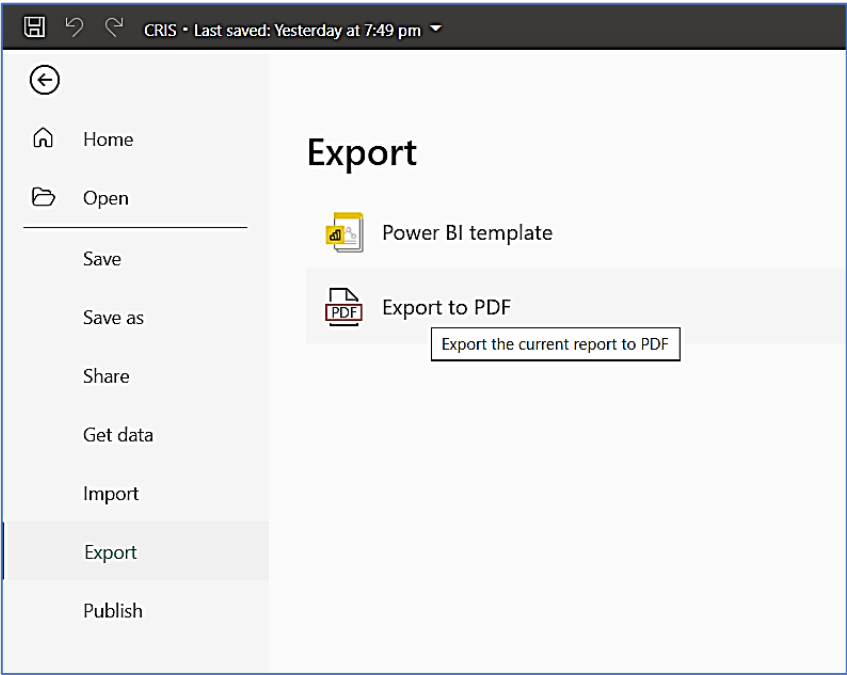


Fig. 9.1 PDF Export Example

Chapter 10: Limitations, Ethics, and Validation

10.1 Limitations of the Project

Although the developed analytical platform for the Crew Management System (CMS) has successfully delivered operational insights through structured dashboards, certain constraints were encountered during its development. One major limitation was the restricted data duration, as the mileage dataset encompassed only five months and was limited to two crew bases/lobbies: TDL and BSP. As a result, broader system-wide analysis across Indian Railways was not feasible within the current scope.

Additionally, manual data integration via Excel sheets restricted the ability to conduct real-time analytics or support automated refreshes, which is vital for live decision-making. The visual design was intentionally oriented towards macro-level trends, which limited detailed drilldowns or individual crew performance views.

While the analytical logic and architecture are scalable, operational deployment in a dynamic production environment would require automated data pipelines, increased geographic coverage, and inter-system data synchronization.

10.2 Ethical Considerations and Data Privacy

The CMS dashboard utilizes sensitive operational and personal data concerning locomotive crew, such as identification codes, risk flags, and medical indicators. During implementation, careful attention was paid to ethical standards and data privacy principles. Personally identifiable information was not displayed explicitly in visual summaries, and the system avoided revealing crew-specific names or direct attributes in public visuals. Aggregate-level summaries were prioritized to maintain confidentiality and compliance.

No external data sources were used, and data processing remained within the analytical framework of Power BI. The project followed a minimum data exposure principle, ensuring that all analytical outcomes were derived without breaching user privacy or violating any institutional data policies.

All datasets were stored securely and used solely for analytical development and visualization within the intended purpose of operational enhancement.

10.3 Feedback and Validation

The final dashboard was reviewed by key personnel associated with CMS operations and supervisory oversight. Constructive feedback was received regarding the layout clarity, KPI consistency, and usability of slicers and visual flow. Modular structure and dynamic interactivity, particularly ability to segment data by designation, base was appreciated.

The integration of operational KPIs such as trip counts, duty hours, and flagged exceptions was well-received as a practical tool for administrative monitoring. Stakeholders noted the potential of this system for scaling across other divisions and requested that future versions include predictive analysis and integration with crew scheduling platforms. The structured presentation of effort distribution, risk profiling, and leave tracking was viewed as a valuable addition to existing CMS workflows.

Chapter 11: Conclusion and Future Work

11.1 Conclusion

This project developed an integrated analytical framework to strengthen supervisory oversight within the Crew Management System (CMS) at CRIS. By consolidating heterogeneous mileage and biodata records from TDL and BSP divisions, static operational datasets were transformed into a structured, modular decision-support platform. Exploratory analytics identified clear structural imbalances between divisions. BSP reflected workload concentration, whereas TDL exhibited substantial latent capacity.

A median-based activation assessment detected 469 underutilized crew members in TDL, translating to nearly 9.08 million kilometres of untapped operational potential. Activation-first redistribution strategies indicated a 22.45% productivity improvement within TDL and a projected 13.75% enhancement in network-level output, without additional manpower or increased operational stress. Workload dispersion also showed potential reduction of approximately 44% through analytical reallocation.

These validated insights were embedded into an interactive Power BI dashboard incorporating engineered KPIs, relational modelling, and dynamic filtering. The solution-maintained scalability, modularity, and supervisory usability while preserving data privacy standards. Overall, the project demonstrates that data-driven workload intelligence can optimize crew deployment and improve operational balance without structural expansion, establishing a scalable foundation for future analytical integration across divisions.

11.2 Future Work

Future enhancements include automating data refresh through direct integration with CMS databases and extending the dashboard across all Indian Railways divisions for broader analysis. Additional capabilities such as predictive crew readiness, anomaly detection, and fatigue risk modelling may be incorporated through machine learning.

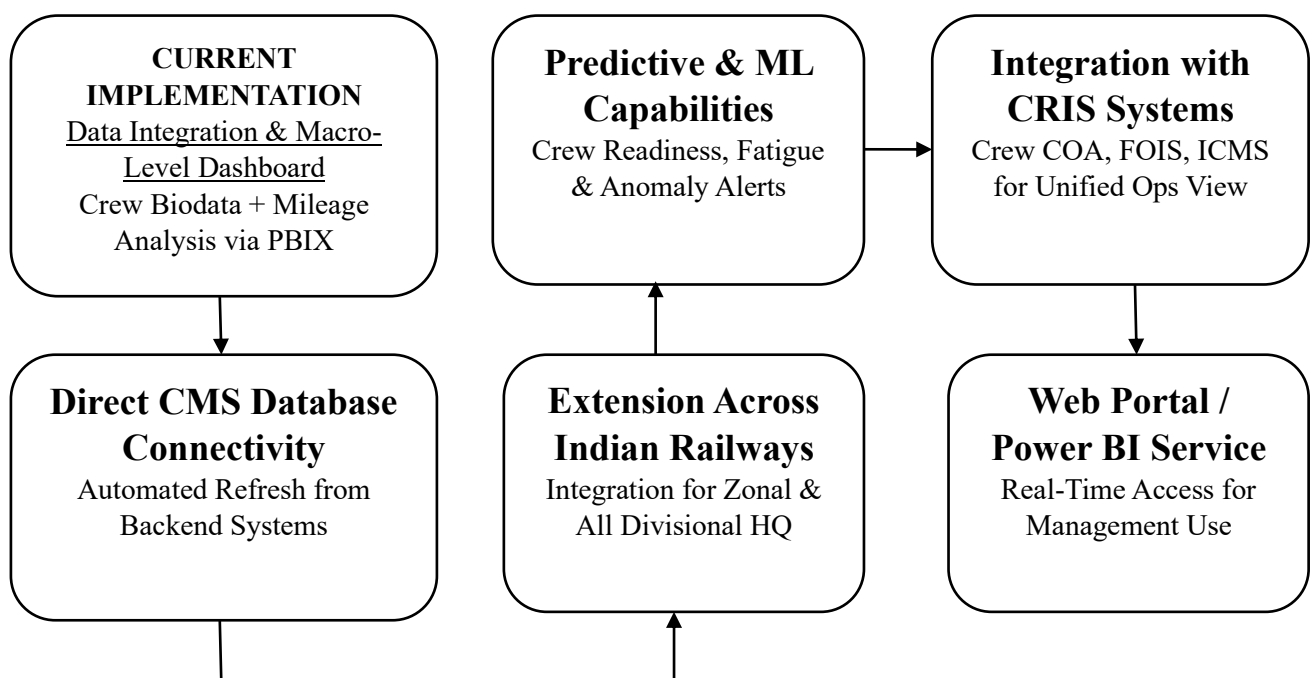


Fig. 11.1 Future Directions

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Appendix A: DAX Measures Used

TABLE A.A.1 DAX MEASURES USED IN THE PROJECT

Measure Name	Description	Purpose
Total Crew	DISTINCTCOUNT(CREW_ID_V)	Counts the number of unique crew members across merged data.
Total_KMs_Run	SUM(TOTAL_KMS)	Calculates the total kilometres run by all crew members.
Total Duty Hours	SUM(TOTAL_DUTY)	Aggregates the total duty hours served.
Total Trips Made	SUM(TRIP_COUNT)	Returns the total number of trips made across all records.
Avg KM per Trip	DIVIDE([Total_KMs_Run], [Total Trips Made])	Computes the average kilometres run per trip.
AvgDutyPerCrew	DIVIDE([Total Duty Hours], [Total Crew])	Shows average duty per crew member.
Avg Trips	DIVIDE([Total Trips Made], [Total Crew])	Calculates average trips made per crew.
RTentative Count	CALCULATE(COUNTROWS(Merged_Mileage), Merged_Mileage[TENTATIVE_FLAG] = "R")	Counts tentative records flagged as Regular.
BTentative Count	CALCULATE(COUNTROWS(Merged_Mileage), Merged_Mileage[TENTATIVE_FLAG] = "B")	Counts tentative records flagged as Backup.
AlcoholCases_ByBase	CALCULATE(COUNTROWS(Biodata), Biodata[ALCOHOL_C] = "Y")	Tallies crew flagged with alcohol cases.
Designation_Count	DISTINCTCOUNT(CREW_DESIG_V)	Counts unique crew designations.
Traction TypeCount	DISTINCTCOUNT(TRCTN_C)	Counts unique traction types.
Total Trips Made	DISTINCTCOUNT(DATE_TIME_D)	Determines number of operational (active) days.
Current_DateTime	NOW() or TODAY()	Displays current system date/time.
LastUpdated	MAX(DATE_TIME_D)	Displays the latest date entry in the dataset for tracking data currency.

Appendix B: Diagrams

This appendix summarizes all the architectural, analytical, and design diagrams used throughout the report. Each diagram provides a visual representation of system flow, data modelling, user interaction, or module architecture, which were essential to understanding and implementing the project logic.

B.1 Data Flow Diagram (DFD - Level 1)

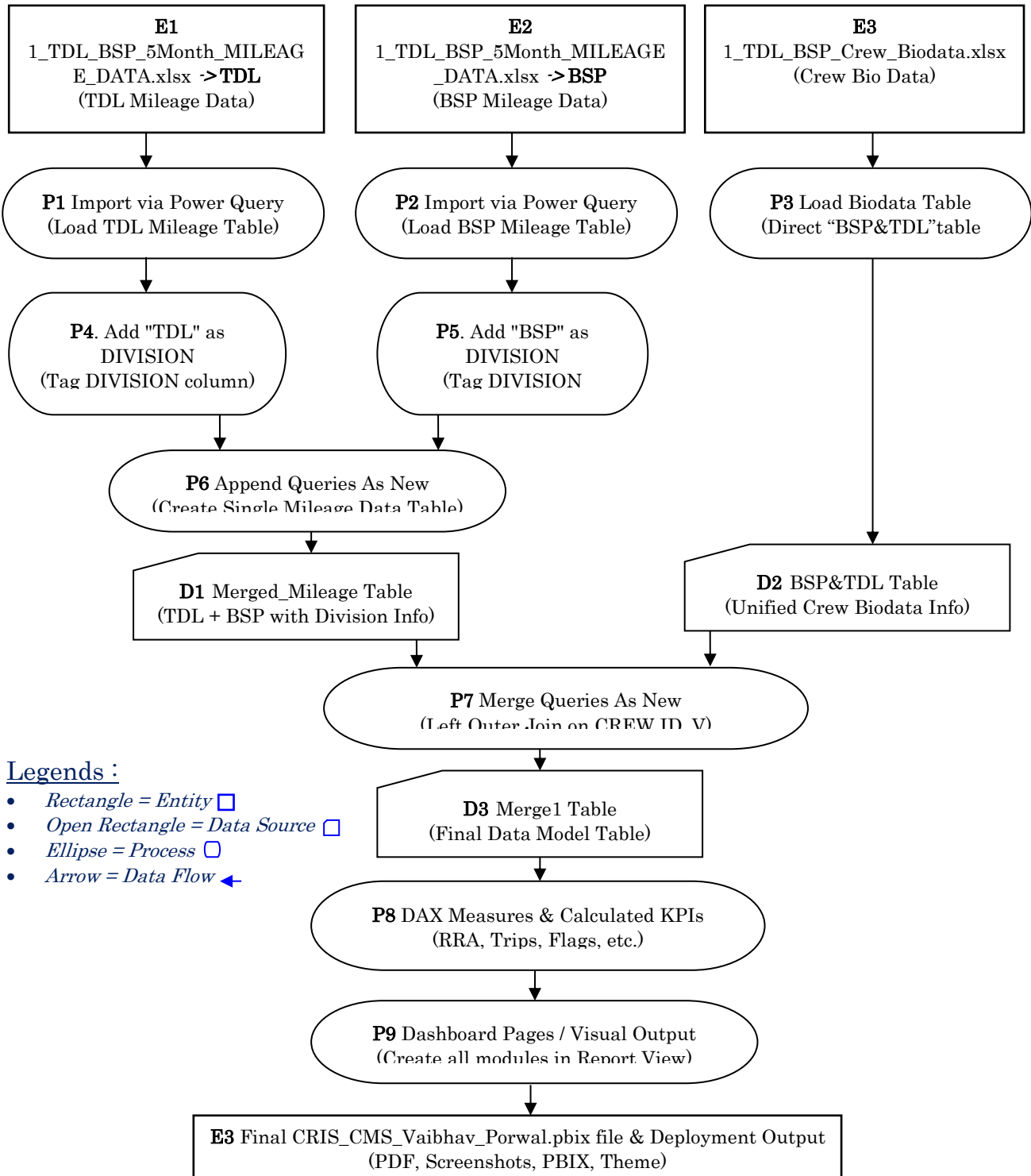


Fig. A.B.1 Data Flow Diagram (DFD - Level 1)

B.2 System Architecture Design

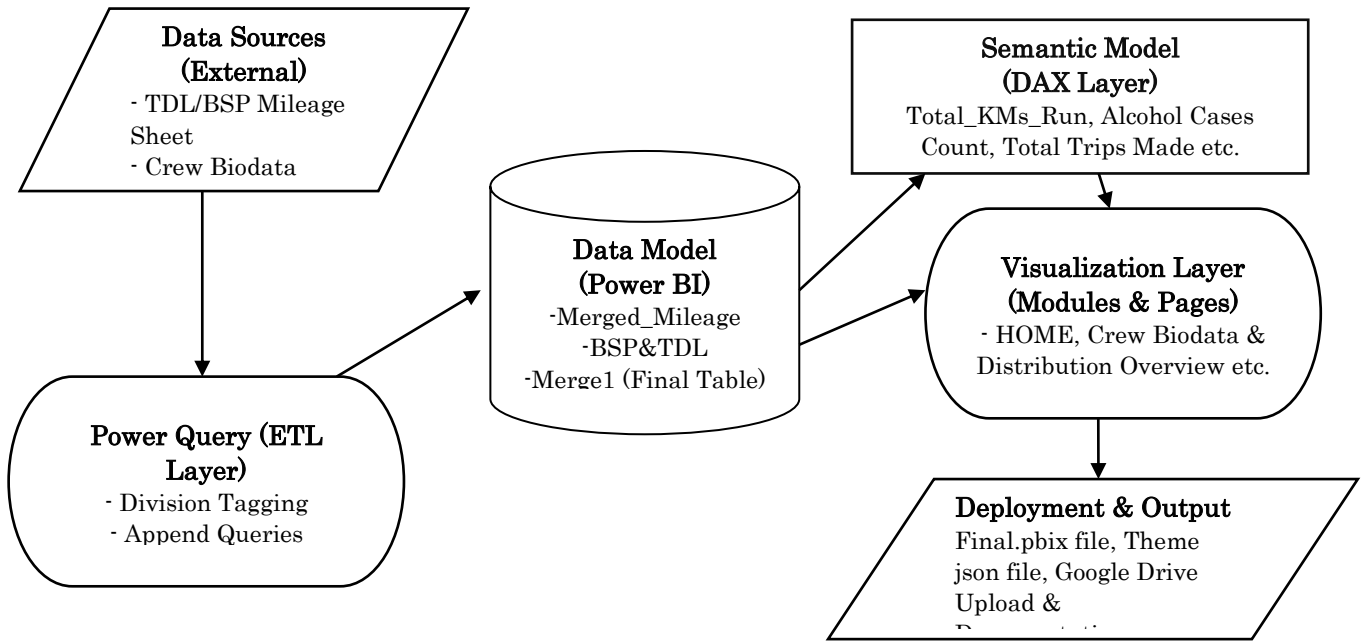


Fig. A.B.2 System Architecture Design

B.3 UML Class Diagram

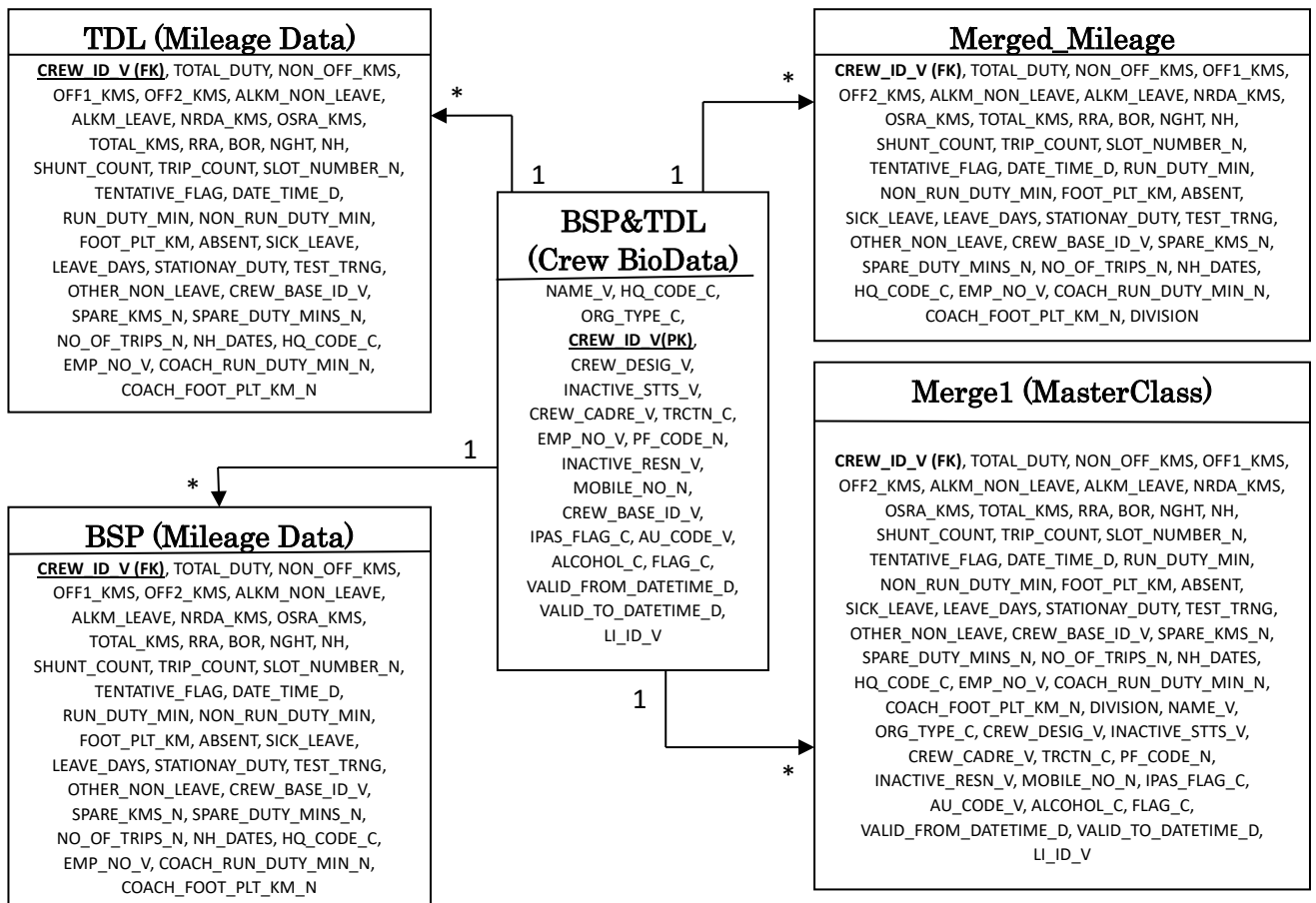


Fig. A.B.3 UML Class Diagram

B.4 PowerBI Model and Schema View

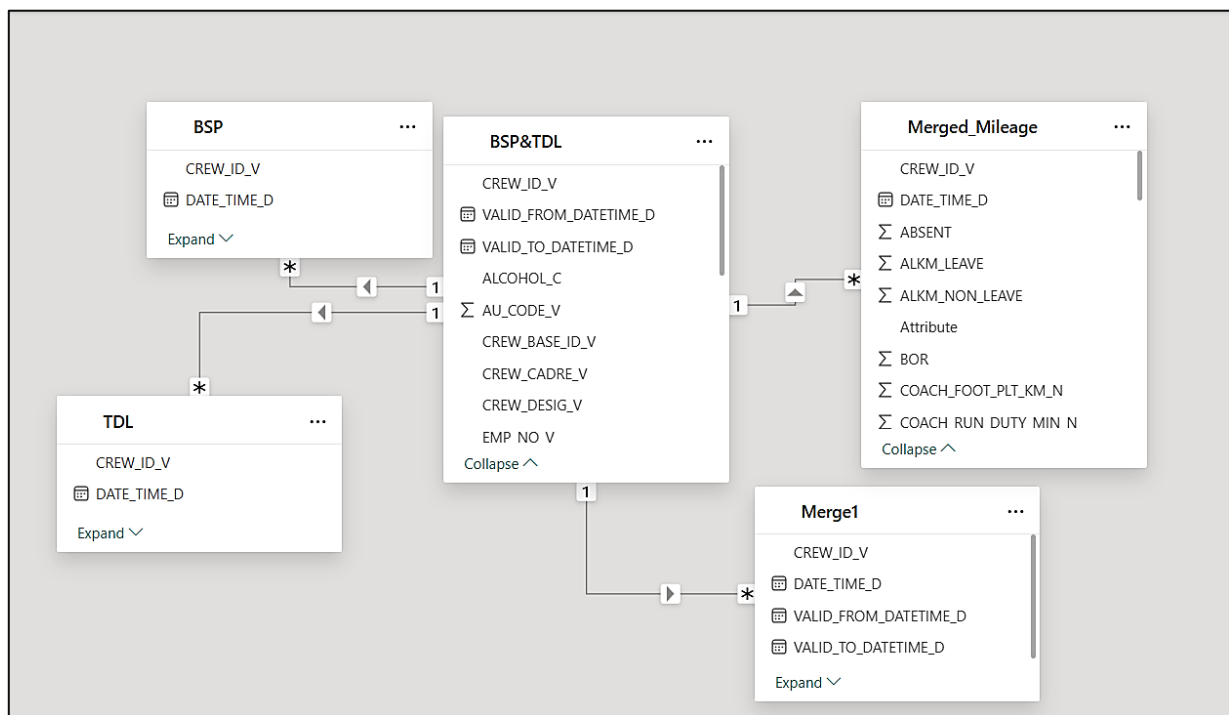


Fig. A.B.4 PowerBI Model and Schema View

B.5 Entity Relationship Diagram

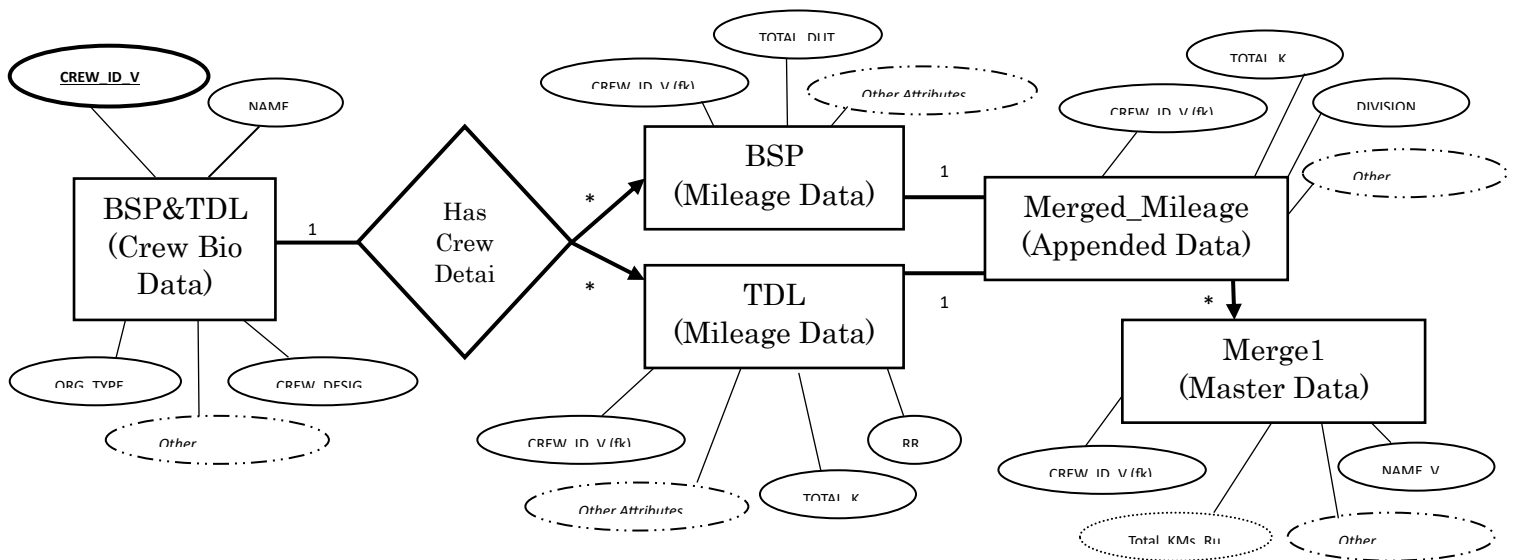


Fig. A.B.5 Entity Relationship Diagram

Appendix C: Framework Modules & Their Functions

TABLE A.C.1 DETAILS ABOUT MODULES USED IN THE FRAMEWORK

Module Name	Key Visuals / Layout	Primary Purpose
Home Page	Summary KPI cards (Total Crew, Trips, KMs), Navigation Tiles	To offer an immediate executive overview of crew deployment and operational status.
Crew Biodata & Distribution Overview	Donut chart (Traction Type), Line chart (Crew Validity by Designation), Pie chart (IPAS Flag), KPIs, slicers	To analyse crew composition by traction type, monitor designation-wise validity trends and IPAS flags.
Distribution and Deployment Metrics	Clustered bar chart, Line chart (Deployment Trends), Matrix table with heatmap, KPI cards	To analyse how duties and deployment volumes vary across designations and bases.
Operational Efficiency & Utilization Trends	Area chart (KMs by Division), Stacked area chart (Trip Trends), Donut chart (ORG_TYPE KMs), Matrix table	To monitor crew workload, trip frequency, and division-wise operational planning trends.
Crew Risk & Operational Efficiency Patterns	Stacked column chart (Inactive Crew), Tree map (Alcohol Flag), Bar chart (Org Type vs Crew Cadre), Donut (RRA)	To monitor risk flags, inactive crew reasons, alcohol cases, and RRA distribution across designations.
Operational Trends & Comparative Insights	Bar chart (Monthly KM Rankings), Scatter (Duty vs Rest vs RRA), Stacked bar (Leave Type Distribution), KPI cards	To correlate rest and duty behavior with leave trends and crew base performance.
Crew Effort Analysis	Bar chart (Stationary & Non-Run Duty), Bar+Line (Total vs Night Duty), Matrix (Run Duty & Coaching), KPI cards	To assess duty load patterns, evaluate night exposure, and monitor coaching effort by designation.
Leave & Traction Utilization	Line chart (Tentative Flag over Time), Stacked bar (Leave vs Non-Leave), Bar chart (Traction Type Trends), KPIs	To evaluate leave trends and traction type deployment for effective planning.